EDEXCEL INTERNATIONAL GCSE MATH (4MA1) • UNIT 2.01

C Algebraic Fractions, Quadratics & Simultaneous Systems

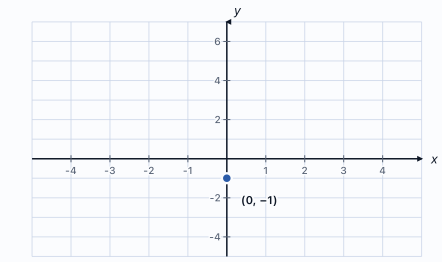
PRACTICE WORKSHEET

25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Algebraic Fractions, Quadratics & Simultaneous Systems Practice Exercises

1 MCQ

The coordinate diagram shows the parabolic curve $f_1(x) = x^2 - 8x + 28$ in the domain of Algebraic Fractions, Quadratics & Simultaneous Systems. By completing the square $f_1(x) = (x - h)^2 + k$, find the sum $k + f_1(6)$.



A 32

B 28

C 16

D 26

2 MCQ

A mathematical model in Algebraic Fractions, Quadratics & Simultaneous Systems is defined by the cubic function $y_2(x) = 3x^3 - 4x^2 + 15x - 9$. Calculate the exact gradient of the tangent to the curve at the point where $x = 3$.

A 96

B 57

C 72

D 34

3 MCQ

In Algebraic Fractions, Quadratics & Simultaneous Systems, two displacement vectors are given by $\mathbf{u}_3 = \langle 4, -8 \rangle$ and $\mathbf{v}_3 = \langle 9, 7 \rangle$. Compute the exact scalar value of $\mathbf{u}_3 \cdot \mathbf{v}_3 + \|\mathbf{u}_3\|^2$.

A -20

B 80

C 172

D 60

4 MCQ

The linear growth profile shown in the diagram for Algebraic Fractions, Quadratics & Simultaneous Systems follows an arithmetic progression with first term $a_1 = 5$ and common difference $d = 6$. Find the exact sum S_{15} of the first 15 terms.

A 705

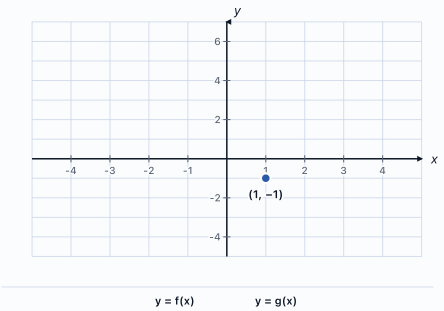
B 89

C 795

D 1335

5 MCQ

In Algebraic Fractions, Quadratics & Simultaneous Systems, evaluate the exact definite integral $I_5 = \int_0^2 (18x^2 + 16x)\,\mathrm{d}x$.



A 104

B 80

C 96

D 41

6 MCQ

In Algebraic Fractions, Quadratics & Simultaneous Systems, consider the quadratic function $f_6(x) = x^2 - 10x + 109$. Express $f_6(x)$ in the completed-square form $(x - h)^2 + k$ and compute the exact value of $k + f_6(12)$.

A 222

B 133

C 217

D 210

7 MCQ

The graph illustrates the cubic model $y_7(x) = 3x^3 - 5x^2 + 33x - 16$ studied in Algebraic Fractions, Quadratics & Simultaneous Systems. Find the instantaneous gradient $\frac{\mathrm{d}y}{\mathrm{d}x}$ of the tangent to the curve at $x = 8$.

A 609

B 496

C 95

D 529

8 MCQ

In Algebraic Fractions, Quadratics & Simultaneous Systems, two displacement vectors are given by $\mathbf{u}_8 = \langle 5, -7 \rangle$ and $\mathbf{v}_8 = \langle 8, 5 \rangle$. Compute the exact scalar value of $\mathbf{u}_8 \cdot \mathbf{v}_8 + \|\mathbf{u}_8\|^2$.

A 79

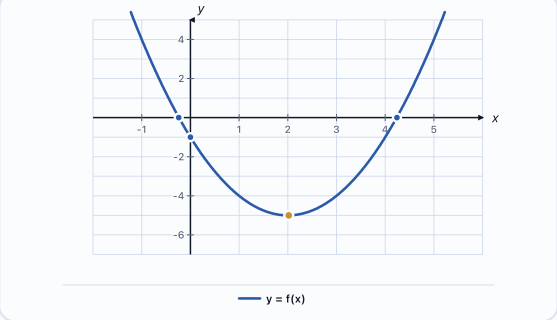
B 5

C 74

D 149

9 MCQ

By completing the square for $y = x^2 - 4x - 1$ (Case #1), find the coordinates of the turning point of the curve.



A $(2, -5)$

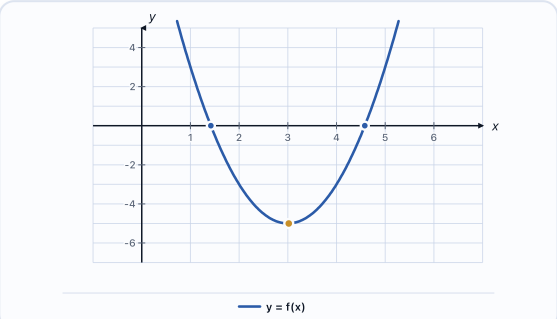
B $(2, -1)$

C $(-2, 5)$

D $(-2, -5)$

10 MCQ

By completing the square for $y = 2x^2 - 12x + 13$ (Case #2), find the coordinates of the turning point of the curve.



A $(-3, 5)$

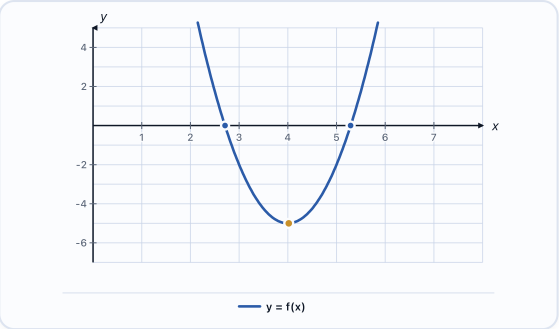
B $(3, 13)$

C $(3, -5)$

D $(-3, -5)$

11 MCQ

By completing the square for $y = 3x^2 - 24x + 43$ (Case #3), find the coordinates of the turning point of the curve.



A $(4, -5)$

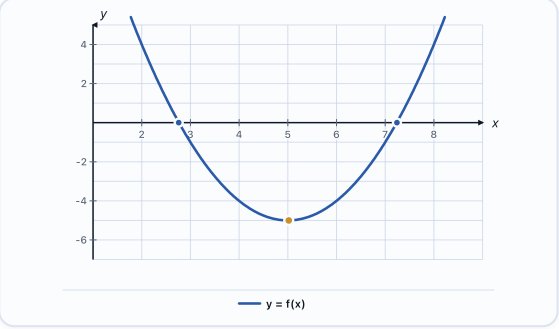
B $(-4, -5)$

C $(-4, 5)$

D $(4, 43)$

12 MCQ

By completing the square for $y = x^2 - 10x + 20$ (Case #4), find the coordinates of the turning point of the curve.



A $(-5, 5)$

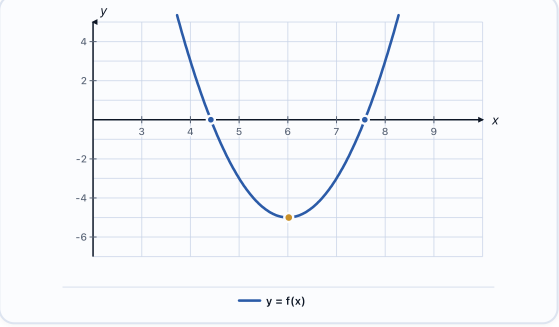
B $(5, -5)$

C $(-5, -5)$

D $(5, 20)$

13 MCQ

By completing the square for $y = 2x^2 - 24x + 67$ (Case #5), find the coordinates of the turning point of the curve.



A $(-6, -5)$

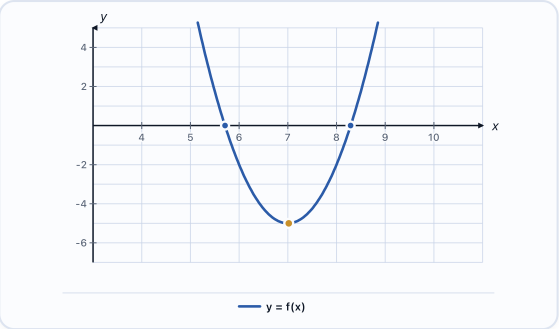
B $(-6, 5)$

C $(6, 67)$

D $(6, -5)$

14 MCQ

By completing the square for $y = 3x^2 - 42x + 142$ (Case #6), find the coordinates of the turning point of the curve.



A $(-7, -5)$

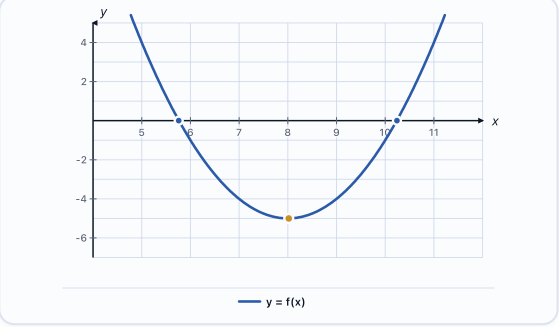
B $(-7, 5)$

C $(7, -5)$

D $(7, 142)$

15 MCQ

By completing the square for $y = x^2 - 16x + 59$ (Case #7), find the coordinates of the turning point of the curve.



A $(-8, 5)$

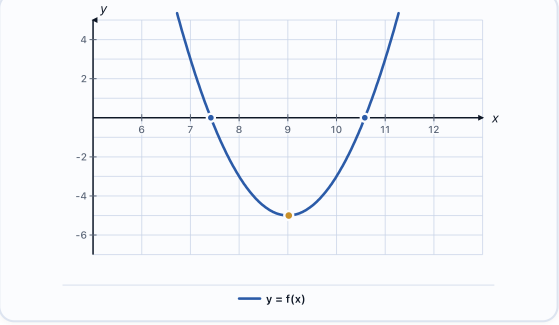
B $(8, 59)$

C $(8, -5)$

D $(-8, -5)$

16 MCQ

By completing the square for $y = 2x^2 - 36x + 157$ (Case #8), find the coordinates of the turning point of the curve.



A $(9, 157)$

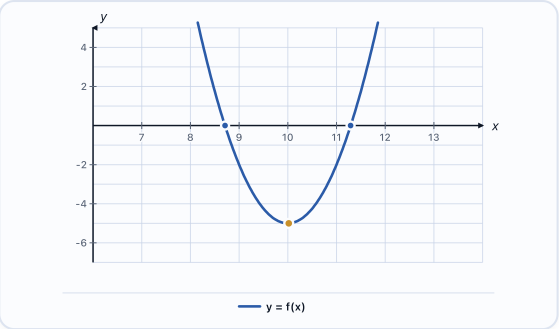
B $(9, -5)$

C $(-9, -5)$

D $(-9, 5)$

17 MCQ

By completing the square for $y = 3x^2 - 60x + 295$ (Case #9), find the coordinates of the turning point of the curve.



A $(-10, -5)$

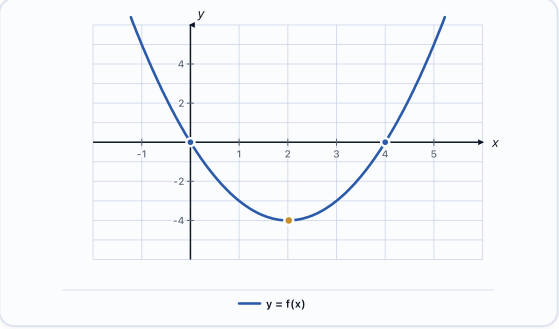
B $(10, 295)$

C $(-10, 5)$

D $(10, -5)$

18 MCQ

By completing the square for $y = x^2 - 4x + 0$ (Case #10), find the coordinates of the turning point of the curve.



A $(-2, -4)$

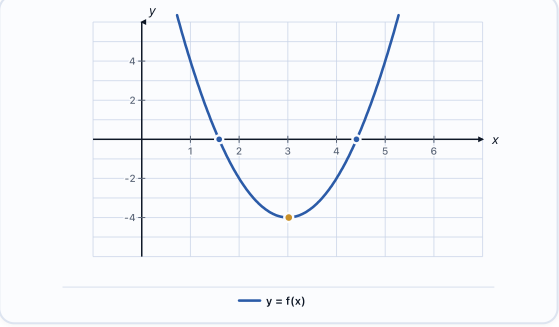
B $(2, -4)$

C $(2, 0)$

D $(-2, 4)$

19 MCQ

By completing the square for $y = 2x^2 - 12x + 14$ (Case #11), find the coordinates of the turning point of the curve.



A $(-3, 4)$

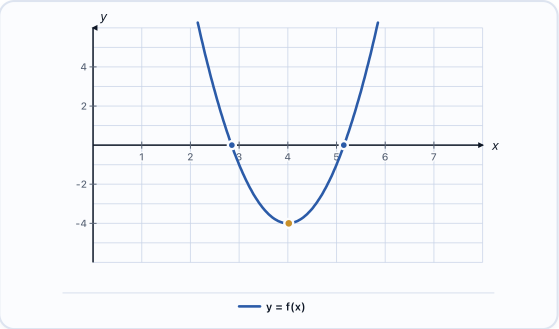
B $(3, -4)$

C $(3, 14)$

D $(-3, -4)$

20 MCQ

By completing the square for $y = 3x^2 - 24x + 44$ (Case #12), find the coordinates of the turning point of the curve.



A $(-4, -4)$

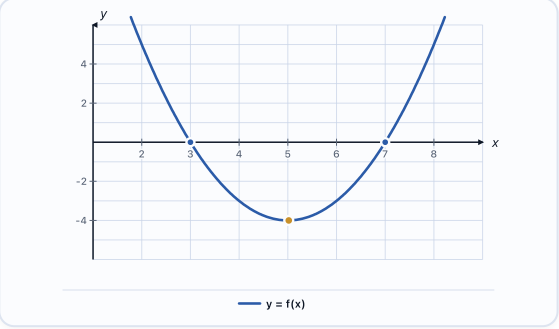
B $(4, -4)$

C $(4, 44)$

D $(-4, 4)$

21 MCQ

By completing the square for $y = x^2 - 10x + 21$ (Case #13), find the coordinates of the turning point of the curve.



A $(-5, -4)$

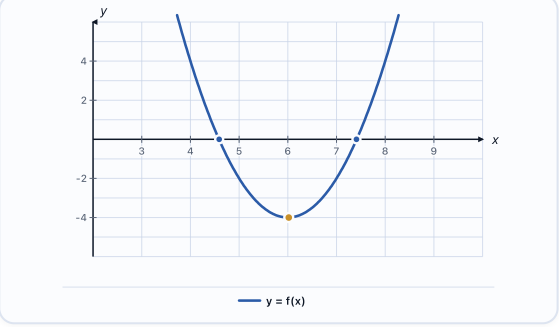
B $(5, 21)$

C $(-5, 4)$

D $(5, -4)$

22 MCQ

By completing the square for $y = 2x^2 - 24x + 68$ (Case #14), find the coordinates of the turning point of the curve.



A $(-6, 4)$

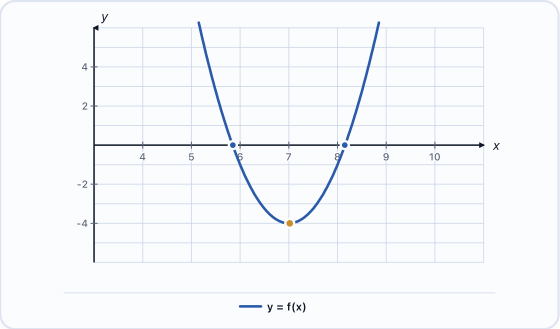
B $(6, 68)$

C $(6, -4)$

D $(-6, -4)$

23 MCQ

By completing the square for $y = 3x^2 - 42x + 143$ (Case #15), find the coordinates of the turning point of the curve.



A $(7, 143)$

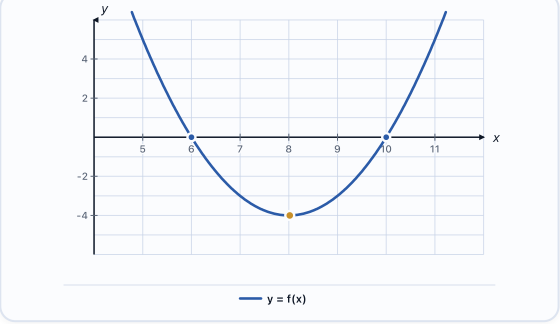
B $(7, -4)$

C $(-7, 4)$

D $(-7, -4)$

24 MCQ

By completing the square for $y = x^2 - 16x + 60$ (Case #16), find the coordinates of the turning point of the curve.



A $(-8, 4)$

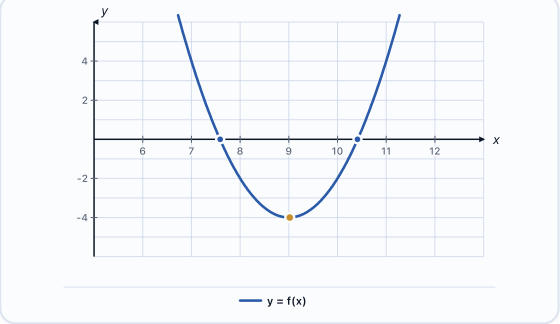
B $(8, -4)$

C $(8, 60)$

D $(-8, -4)$

25 MCQ

By completing the square for $y = 2x^2 - 36x + 158$ (Case #17), find the coordinates of the turning point of the curve.



A $(-9, 4)$

B $(9, -4)$

C $(9, 158)$

D $(-9, -4)$